

exists, or is coming from an ISO file. The entire virtual machine could be moved to another folder, drive or even computer and it would just work. You could even duplicate it. You might replace hardware components in the host computer and in the end nothing changes for the virtual machine. In the case of Windows Storage Spaces, you can work with your files and folders without worrying about the physical location of the data, or which hard drive it is stored on. The hard drives can be replaced as they get old or run out of space, but you won't need to care, all that matters is the data.

Now that we have an understanding of the distinction between emulation and virtualisation, what do we make of the term simulation. As we did for virtualisation, let's look at examples of simulation.

Simulation

A flight simulator is a tool for creating an environment that closely mimics that of an actual aeroplane in flight, and during take-off and landing. There are numerous flight simulator games that mimic certain aspects of the experience of flying a plane, but there are also intricate mechanisms that simulate the entire cockpit of an aeroplane and allow you to operate a simulated air plane just as you would a real one.

The goal of such a simulation is to make the interaction of the pilot with the airplane and the interaction of the plane with its environment as authentic as possible. As far as the pilot is concerned they are in a real plane. The simulation then allows one to test the pilot in conditions where parts of the airplane fail without endangering anyone.

Another popular use of simulations is in science, to predict weather, run protein folding scenarios, planetary motion, fluid dynamics etc. In all these cases the underlying system behind these systems is built in software with the intent of testing the outcome of certain scenarios without performing them for real. This could be because real testing is impossible, infeasible or just expensive.

A very popular example of a simulator, and one that is similar in function and purpose to a virtual machine / emulator is the iOS simulator. The iOS simulator is an application that is available for free as part of the iOS SDK and XCode on Apple's OSX. It has a role similar to the Android emulator and Windows Phone emulator as it allows you to test applications developed for iOS without a physical device. Why is the iOS simulator a simulator then and not an emulator?